Comment on "Band topology of the breathing kagome lattice"

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(Dated: September 15, 2026)

Geschner *et al.* [Phys. Rev. B **113**, 155151 (2026)] reported that, at vanishing intracell hopping, only 12 state energies near each corner change during their local interpolation of nearest-neighbor hoppings. Exact reduction of the raster-defined deformation instead gives a 15-site invariant corner component with characteristic polynomial $\chi(E,\lambda) = (E-1)^2 P_6(E,\lambda)P_7(E,\lambda)$. The complete λ-independent factor is $(E-1)^2$, so 13 algebraic eigenvalue roots per corner are nonconstant. The three disjoint components in the 630-site flake therefore contribute 39, rather than 36, nonconstant roots. This corrects the finite spectral count but does not change the original paper's conclusion that the zero-energy corner state can be removed.

I. SCOPE OF THE CORRECTION

The nearest-neighbor deformation introduced by Geschner *et al.* removes the zero-energy corner states of a breathing-kagome flake without breaking the relevant crystalline or tripartite constraints [1]. At $t = 0$, where t and t' denote the intra- and intercell hopping amplitudes, respectively, the authors state that only 12 state energies near a corner vary with their interpolation parameter λ. The same number appears in the author preprint and the published article. Here I give an exact component calculation showing that the number is 13 when roots are counted with algebraic multiplicity. This counting issue is separate from the broader discussion of corner-state topology and destructive-interference protection [2], and no change to the physical conclusion of Ref. [1] is implied.

Throughout, $t = 0$, $t' = 1$, and a positive bond weight w_{ij} enters the single-particle Hamiltonian as $H_{ij} = -w_{ij}$. The interpolation is

$$H_\lambda = (1-\lambda)H + \lambda H', \qquad 0 \le \lambda \le 1. \tag{1}$$

All statements in the central calculation refer to the bond pattern rendered in Fig. 3 of Ref. [1]; they do not assert a result for an arbitrary corner perturbation.

II. EXACT CORNER COMPONENT

At $t = 0$, the union of the initial and final bonds affected near one corner has a connected 15-vertex component. Labeling its vertices $0,\dots,14$ as in Fig. 1, its 17 edges separate into

$$\begin{aligned}
\mathcal{E}_\lambda &= \{(0,1),(0,2),(1,2),(3,4),\\
&\qquad (5,7),(6,8),(11,13)\},\\
\mathcal{E}_{1-\lambda} &= \{(1,3),(2,5),(4,6),(7,11),\\
&\qquad (8,9),(8,12),(10,13),(13,14)\},\\
\mathcal{E}_1 &= \{(9,12),(10,14)\}.
\end{aligned} \tag{2}$$

The ordered pairs denote undirected edges; the three sets carry weights λ, $1-\lambda$, and 1. Equation (2) is therefore a complete specification of the finite matrix used below.

The corner-bisector reflection acts as

$$(1\,2)(3\,5)(4\,7)(6\,11)(8\,13)(9\,10)(12\,14), \tag{3}$$

with vertex 0 fixed. Restriction to the mirror-even and mirror-odd spaces, of dimensions 8 and 7, gives exact characteristic polynomials

$$\begin{aligned}
\chi_+(E,\lambda) &= (E-1)P_7(E,\lambda),\\
\chi_-(E,\lambda) &= (E-1)P_6(E,\lambda),
\end{aligned} \tag{4}$$

where the subscripts on P_7 and P_6 denote their degrees in E. Thus

$$\chi(E,\lambda) = \det(EI - H_\lambda) = (E-1)^2 P_6(E,\lambda)P_7(E,\lambda). \tag{5}$$

The endpoint factorizations supply a transparent first check. At $\lambda = 0$ the component is an isolated site, four dimers, and two triangles, whereas at $\lambda = 1$ it is six dimers and one triangle. Hence

$$\chi(E,0) = E(E-1)^8(E+1)^4(E+2)^2, \tag{6}$$

$$\chi(E,1) = (E-1)^8(E+1)^6(E+2). \tag{7}$$

It remains to establish that Eq. (5) contains the *complete* interpolation-independent factor. Write $\chi(E,\lambda) = \sum_{m=0}^{12} c_m(E)\lambda^m$. Exact fraction-free elimination gives

$$\gcd_{0\le m\le 12} c_m(E) = (E-1)^2. \tag{8}$$

Maximality can be checked from only two coefficients. After division by $(E-1)^2$, the coefficients of λ^{11} and λ^{12} are

$$q_{11}(E) = 50E^2 - 196E - 22, \qquad q_{12}(E) = 24E + 4. \tag{9}$$

The sole root of q_{12} is $-1/6$, while $q_{11}(-1/6) = 217/18$, so q_{11} and q_{12} are coprime. No additional polynomial in E can therefore divide every c_m.

A λ-independent energy would contribute a factor of $E - E_0$ to every coefficient of χ. Equations (8) and (9)

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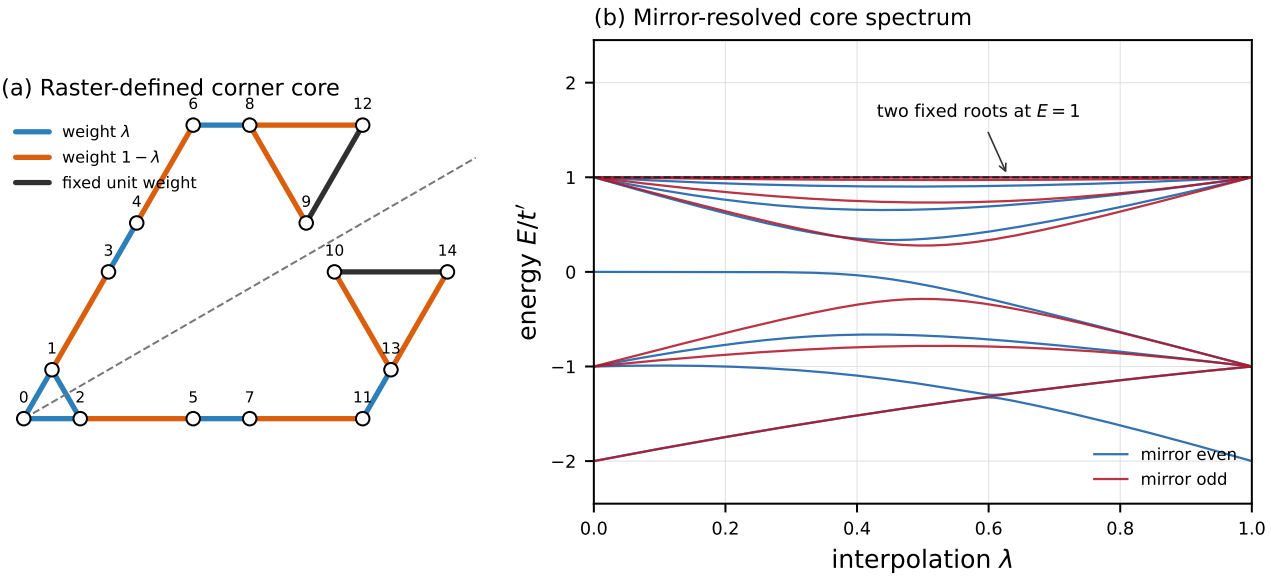


FIG. 1. (a) The complete 15-site invariant component specified by Eq. (2). The dashed line is the reflection axis. (b) Spectrum obtained by diagonalizing the two reflection sectors; one $E = 1$ root in each sector is fixed, while the remaining 13 roots are nonconstant. Lines serve only as a visualization; the count follows from the exact polynomial argument in the text. The vector-figure script was prepared with OpenAI Codex assistance and reads the exact edge certificate; the plotted eigenvalues were checked against the exact endpoint and factorization data.

consequently show that precisely two roots, counted with generic algebraic multiplicity, are fixed throughout the interpolation. The remaining $15 - 2 = 13$ roots are nonconstant. This statement does not mean that 13 distinct curves must be visually resolvable at every λ: degeneracies with the flat edge and bulk bands can conceal a moving algebraic root.

In the $N = 20$ complete-cell triangular flake (630 sites), exact lattice rotation gives three pairwise-disjoint copies of the component in Eq. (2). They contain all 21 bonds ramped on and all 24 bonds ramped off; the exterior block is independent of λ at $t = 0$. The nonconstant factor of the three-corner characteristic polynomial therefore has degree

$$3 \times 13 = 39, \tag{10}$$

rather than the $3 \times 12 = 36$ inferred from the published count.

III. CHECKS AT FINITE INTRACELL HOPPING

The bivariate polynomial was obtained in exact integer/rational arithmetic from the graph in Eq. (2), both directly and after mirror reduction. A separate implementation, sharing no polynomial code, evaluated the characteristic polynomial by the Faddeev–LeVerrier recurrence at 16 integer values of λ and reconstructed all coefficients by Newton interpolation. Two unused rational values, $\lambda = 1/2$ and $-2/3$, agree exactly. As another check, at $\lambda = 1/2$ one finds $\operatorname{rank}(H_\lambda - I) = 13$, $\chi(-1, 1/2) = 27/256$, and $\chi(-2, 1/2) = -1112607/256$. These evaluations independently exclude a third fixed root among the energies common to the two endpoint spectra.

For context, the finite-t calculation also sharpens how this local rearrangement meets the edge spectrum. On the balanced N-cell-side termination, let $C = N(N + 1)/2$. For $N \geq 6$, a square submatrix of the $A \to B$ outgoing block is triangular and has exact determinant

$$\det H_{BA}(\lambda) = (-1)^C t^{C-7}\big[(1 - \lambda)t + \lambda t'\big]^7. \tag{11}$$

The cyclic partners obey the same formula. Thus at $t = 0.1$, $t' = 1$, the finite balanced flake has no exact eigenstate supported on only one sublattice, even though its exponentially small boundary residual can mimic a numerical kernel.

For $N = 20$, exact rational inertia in the mirror-even A_1 and mirror-odd A_2 sectors proves an order reversal of the 44th A_1 and 33rd A_2 levels (zero-based indices 43 and 32) between

$$\lambda_L = \frac{7919395246780079}{10^{16}}, \qquad \lambda_R = \frac{3959697623410039}{5 \times 10^{15}}. \tag{12}$$

Continuity and opposite mirror parity therefore imply at least one exact crossing in this interval, near $E = -0.554930413681\, t'$. This is a finite-flake statement; neither crossing uniqueness nor thermodynamic persistence follows.

The uniform edge outside the deformed core admits an evanescent ansatz $(A, B, C)_{i,j} = r^i z^j (a, b, 0)$. Direct substitution gives

$$E^2 = t^2 + t'^2 + tt'(r + r^{-1}),$$
$$\frac{b}{a} = -\frac{t + t'r}{E}, \qquad z = -\frac{t(1 + b/a)}{t'[1 + (b/a)/r]}. \tag{13}$$

At the quoted crossing energy, the decaying solution has $r = -0.1454530797104283$ and $z = -0.0587345561172001$. The negative r accounts for the observed alternating finite-size approach. Its numerical agreement with finite-flake ratios is evidence for the mechanism, not a proof of a unique infinite-size secular root.

IV. CONCLUSION

The local deformation of Ref. [1] affects 13 nonconstant algebraic roots per isolated corner component at $t = 0$, not 12. The correction follows from a fully specified 15-site graph and an exact maximal common-factor calculation. It changes neither the removal of the corner zero mode nor the obstructed-atomic-limit conclusion of the original paper. It also supplies no general connectivity criterion, no bulk invariant, and no thermodynamic phase-transition claim.

DATA AVAILABILITY

The exact graph and polynomial certificate, two independent reconstruction scripts, the finite-crossing certificate, the edge-transfer data, the plotting source, and environment specifications are included as ancillary files in the submission package.

ACKNOWLEDGMENTS

OpenAI Codex (model GPT-5.6-sol) assisted in a user-directed workflow with primary-source comparison, exact-algebra and Python-code development, independent computational checks, vector-figure preparation, and manuscript drafting. The central calculation was subjected to the independent exact procedures described above. The AI system is not an author. Before any submission, the named author must review the sources, reproduce and assess the calculations, verify the figure and text, confirm permissions and the disclosure, and take responsibility for the final manuscript.

[1] C. K. Geschner, A. Y. Chaou, V. Dwivedi, and P. W. Brouwer, Band topology of the breathing kagome lattice, Physical Review B **113**, 155151 (2026).

[2] M. A. J. Herrera, S. N. Kempkes, M. Blanco de Paz, A. García-Etxarri, I. Swart, C. Morais Smith, and D. Bercioux, Corner modes of the breathing kagome lattice: Origin and robustness, Physical Review B **105**, 085411 (2022).